National University
Of Computer and Emerging Sciences

Foundation of Advancement
Of Science and Technology

Lab Number:	08
Course Code:	CL2001
Course Title:	Data Structures - Lab
Semester:	Fall 2026
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Guidelines

- You may use **only the concepts covered in the lab** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
  Example: *25p-1234_ozair_labtask07.pdf*
- Unethical use of AI tools will result in **0 marks**.
- Late submissions will incur a penalty of **-3 marks per day**.

Tasks

NOTE: USE OF GLOBAL OR STATIC VARIABLES IS NOT ALLOWED

1 Recursive Multiplication Table

Write a recursive function that prints the multiplication table of a given number `m` from 1 up to a specified limit `n`.

Example:

```
Input: m = 2, n = 10
Output:
2x1=2
2x2=4
2x3=6
...
```

2 Recursive Power Calculator

Write a recursive function that calculates the power of a number given its base `b` and exponent `p`. The function should return the calculated result.

Example:

```
Input: b = 2, p = 5
```

```
Output: Ans = 32
```

3 Count Odd Values in a Linked List

Write a recursive member function that counts and returns the number of odd-valued elements in a linked list.

Example:

```
Input: 11 -> 22 -> 33 -> 41 -> 55 -> End
Output: OddCount = 4
```

4 Search in a Linked List

Write a recursive member function that searches for a given value in a linked list. The function should return `true` if the value exists in the list and `false` otherwise.

Example 1:

```
Input:
List: 4 -> 2 -> 3 -> 1 -> End
SearchItem: 2

Output: True
```

Example 2:

```
List: 4 -> 2 -> 3 -> 1 -> End
SearchItem: 10

Output: False
```

5 Mean of a Linked List

Write a recursive member function that calculates and returns the mean (**average**) of the values stored in a linked list.

The mean is calculated by dividing the sum of all the elements by the total number of elements. The function should both calculate and return the average as a numeric value.

Example 1:

```
Input: 1 -> 2 -> 3 -> 4 -> End
Output: Average = 2.5
```

Example 2:

```
Input: 11 -> 22 -> 33 -> 41 -> 55 -> End
Output: Average = 32.4
```